that is a straightforward No), but “Can a collapsing universe bounce back and start again?”

Cyclic universes that go from Bang to Crunch and back again forever have a certain appeal in their tidiness. (And we’ll explore these in more detail in Chapter 7.) Rather than a beginning from nothing and catastrophic, final end, a cycling universe can in principle bounce along in time arbitrarily far in each direction, with endless recycling and no waste.

Of course, like everything in the universe, it turns out to be significantly more complicated. Based purely on Einstein’s theory of gravity, general relativity, any universe with a sufficient amount of matter has a set trajectory. It starts with a singularity (an infinitely dense state of spacetime) and ends with a singularity. There isn’t really a mechanism in general relativity to transition from an end-singularity to a beginning one, however. And there is reason to believe that none of our physical theories, general relativity included, can describe the conditions of anything close to that kind of density. We have a pretty good understanding of how gravity works on large scales, and for relatively (ha!) weak gravitational fields, but we have no idea how it works on extremely small scales. And the kinds of field strengths you’d encounter when the entire observable universe is collapsing into a subatomic dot are all *kinds* of incalculable. We can be fairly confident that for that particular situation, quantum mechanics should become important and do *something* to make a mess of things, but we honestly don’t know what.

Another problem with a bouncing Crunch-Bang universe is the question of what makes it through the bounce. Does anything survive from one cycle to another? The asymmetry I mentioned between an expanding young universe and a collapsing old one, in terms of the radiation field, is actually potentially very problematic here, as it implies that the universe gets (in a precise, physically meaningful sense) messier with every cycle. That makes the cyclic universe less appealing from the standpoint of some very important physical principles that we’ll discuss in later chapters, and it’s certainly more difficult to fit into a nice neat reduce-reuse-recycle ecology.

THE ALLURE OF THE INVISIBLE

Bounce or no bounce, a universe with too much matter and not enough expansion is destined for a crunch, so checking where we’re at in terms of